

COMP 6704: Advanced Topics in Optimization

Group Project

PolyU Fall 2026

1 Group Formation

Please form teams consisting of 4 members for this project. If you wish to work in teams with more than 4 members, you must seek approval by sending an email to the instructor along with a valid justification. Please form your team by selecting a group in Canvas under **People/Group Project** by **13 September**. Failure to submit your team information by this deadline will result in random team assignments. To ensure you have sufficient time to finish this project, any change of your team will **NOT BE ALLOWED after 13 September**. If you have any concerns, please reach out to the teaching team as early as possible.

2 Project Task

Following the group formation, both the project topics and the presentation sequence will be assigned through a random lottery, and the results will be announced on **15 September**. Commencing in Week 7, four groups will present their projects each week.

Table 1: Group Project Topics.

Week	Date	Project Topic
7	13 October	Linear Programming
8	20 October	Linear Programming
9	27 October	Convex Optimization
10	3 November	Convex Optimization
11	10 November	Unconstrained Optimization
12	17 November	Unconstrained Optimization
13	24 November	Constrained Optimization

The specific topic assigned for each week is outlined in Table 1. For each appointed project topic, you may choose to focus on a particular optimization problem by conducting a comprehensive literature review, implementing various optimization algorithms using platforms such as Python or MATLAB, and performing experimental comparisons of their performance. Finally, you should summarize the strengths and weaknesses of the different algorithms.

2.1 Candidate Project Topics

The following are suggested problems for each optimization topic:

- **Linear Programming:** Supply Chain Network Optimization, Resource Allocation in Cloud Computing, Transportation and Logistics Optimization.
- **Convex Optimization:** Machine Learning Model Training via Convex Optimization, Signal Processing and Compressed Sensing, Network Flow and Communication Optimization.
- **Unconstrained Optimization:** Deep Learning Optimizer Comparison, Hyperparameter Optimization and Bayesian Methods, Financial Portfolio Risk Minimization.
- **Constrained Optimization:** Autonomous Vehicle Path Planning, Energy Management in Smart Grids, Multi-objective Engineering Design Optimization.

These topics serve as starting points for your projects. You are encouraged to customize and specialize these topics based on your group's interests and expertise while ensuring they align with the fundamental concepts of optimization theory covered in the course.

3 Submission Guideline

Submission Deadline: 11:59 PM on Sunday of the week in which your group presents. Late submissions will incur a 20 scores deduction from the total score (100 scores) for each day past the deadline.

Project Report (page limit: up to 15 pages): The project report should include the following key components: Introduction, Methodology, Experimental Results and Analysis, and Conclusions. Given that the selected problem may be addressed using various optimization algorithms, you are expected to implement multiple algorithms and compare their performance in terms of criteria such as convergence rate and stability. As a guideline, each team member should independently implement at least one algorithm related to the chosen optimization problem. You are encouraged to include visualizations to clearly illustrate the comparative results and highlight the distinctive characteristics of each algorithm. Please ensure that the **names and student IDs of all team members** are clearly listed at the beginning of your report. You are encouraged to use the [NeurIPS 2025 template](#) to write your report.

Group Presentation (time requirement: 30 minutes presentation + 10 minutes Q&A): Please prepare a PPT presentation that highlights the key components of your project report. When organizing the content of your presentation, it is recommended to follow the same structure as the Project Report for consistency and clarity. **Each group member is required to present and answer questions.**

Submission Rule: Please combine your final project report and presentation slides into a single PDF file and submit it to Canvas. Please name the file with your Group ID, such as "Group10.pdf". The final PDF file should be submitted to **Canvas** and the submission entry is: Assignments/Group Project. Please keep the size of the uploaded file as small as possible (less than 40 MB). Take note that only one team member needs to upload the final file to Canvas. Don't forget to double-check if the submission is saved successfully before leaving. Multiple submissions are allowed, and we will only mark the last submission.

Extra Note Engaging in any form of cheating or plagiarism, including, but not limited to, copying assignments from previous years, will result in a score of zero for this project. All individuals involved in such misconduct will be reported to the relevant department and the Graduate School for further disciplinary action.

4 Grading Rubrics

The group project will be evaluated using the following metrics. Each project will receive a single grade, which will be assigned to all team members.

- **Project Report (50%):** This component assesses the technical rigor and academic quality of your written submission. Evaluation will consider the thoroughness of the background survey, clarity of problem formulation, effectiveness of algorithmic implementation, depth of experimental analysis, and the reproducibility of your code.
- **Presentation (40%):** This component assesses your ability to effectively communicate your research findings through oral presentation, including language proficiency, structural organization, time management, and depth of responses to questions.
- **Attendance (10%):** This component evaluates your participation and engagement in class presentations, including attendance at other groups' presentations and active involvement in the Q&A sessions.

The details marking criteria are summarized in Table 2.

Table 2: Group Project Evaluation Metrics.

Category	Metric	Weightage	Description
Report	Background Survey	10%	A comprehensive literature review that demonstrates a good understanding of related work and establishes the theoretical foundation for your project is essential. Please make sure your selected optimization problem falls into the topic of the week.
Report	Problem Formulation and Motivation	10%	A mathematically rigorous formulation of the optimization problem with clear objective functions, constraints, and variables. The motivation should justify why this problem is worth studying and its relevance to real-world applications.

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Table 2 – continued from previous page

Category	Metric	Weightage	Description
Report	Algorithm Design and Implementation	10%	The sophistication of proposed or implemented algorithms, diversity of comparison methods, and algorithmic innovations. This includes theoretical analysis of computational complexity, convergence properties, and algorithmic contributions.
Report	Experimental Design and Analysis	10%	Comprehensive experimental evaluation including performance metrics (convergence rate, solution quality, computational efficiency), statistical analysis, visualization of results, and comparison with baseline methods. Ablation studies and sensitivity analysis are highly valued.
Report	Code Quality and Reproducibility	10%	Well-documented, readable, and reproducible code with clear instructions for replication. GitHub repository link must be included in the report (either in the abstract or as a footnote). Please make sure you have included everything that allows to reproduce all experimental results presented in the report. A comprehensive README documentation is crucial for collaborative projects and long-term maintenance.
Presentation	Language Proficiency	10%	Clear pronunciation, appropriate vocabulary, grammatical accuracy, and professional delivery. Effective use of technical terminology and ability to communicate complex concepts in accessible language.

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Category	Metric	Weightage	Description
Presentation	Structure & Flow	10%	Logical organization of content with smooth transitions between sections. Clear introduction, methodology, results, and conclusions. Effective use of visual aids and adherence to standard academic presentation format.
Presentation	Time Management	10%	Adherence to the allocated 30-minute presentation time with balanced coverage of all sections. Appropriate pacing that allows for adequate explanation of key concepts without rushing or excessive delays.
Presentation	Response Quality and Depth	10%	Thoughtful and knowledgeable responses to questions during the Q&A session. Demonstration of deep understanding through ability to address follow-up questions, acknowledge limitations, and suggest future research directions.
Presentation	Attendance	10%	Regular attendance at class presentations and active participation in Q&A sessions for other groups' presentations. After each group completes their presentation, four groups will be randomly selected to answer questions about the presented work. If no one is present or no questions are asked, points will be deducted accordingly.

5 Consultation

You are welcome to send any questions regarding the group project to the teaching assistant, Mr. Yuhui Wu (yuhui.wu@connect.polyu.hk). It is strongly recommended that your email subject line begins with “COMP6704+Group Project+Group id” (for example, “COMP6704+Group Project+Group5”). If you do not receive a reply within 48 hours, please forward your query to the course instructor.